

CLASS X PRACTICE QUESTION PAPER

CBSE-STYLE PRACTICE EXAMINATION

SUBJECT: MATHEMATICS (STANDARD)

Time Allowed: 3 Hours

Maximum Marks: 100

General Instructions:

1. This question paper contains three sections: Section A, Section B, and Section C.
 2. Section A consists of Multiple Choice Questions (MCQs).
 3. Section B consists of Short Answer Type Questions.
 4. Section C consists of Long Answer Type Questions.
 5. All questions are compulsory. Marks for each question are indicated on the right.
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SECTION A - MULTIPLE CHOICE QUESTIONS

(36 Questions * 40 Marks)

- Q1.** If the perimeter and the area of a circle are numerically equal, then the radius of the circle is: [1]
(A) 2 units
(B) π units
(C) 4 units
(D) 7 units
- Q2.** The area of a quadrant of a circle whose circumference is 22 cm is (use $\pi = 22/7$): [1]
(A) $77/8 \text{ cm}^2$
(B) $77/4 \text{ cm}^2$
(C) $77/2 \text{ cm}^2$
(D) 154 cm^2
- Q3.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes (use $\pi = 22/7$). [2]
(A) $154/3 \text{ cm}^2$
(B) $154/6 \text{ cm}^2$
(C) $308/3 \text{ cm}^2$
(D) $77/3 \text{ cm}^2$
- Q4.** The 11th term of the AP: -3, $-1/2$, 2, ... is: [1]
(A) 28
(B) 22
(C) -38
(D) $-46 \frac{1}{2}$
- Q5.** The 10th term of the AP: 2, 7, 12, ... is: [1]
(A) 47
(B) 49
(C) 52
(D) 45
- Q6.** The sum of all two-digit odd positive integers is: [2]
(A) 2475
(B) 2530
(C) 4905
(D) 5049
- Q7.** From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is: [1]
(A) 7 cm
(B) 12 cm
(C) 15 cm
(D) 5 cm

- Q8.** A line intersecting a circle in two points is called a: [1]
(A) Tangent
(B) Secant
(C) Chord
(D) Diameter
- Q9.** Find the coordinates of the point which divides the join of $(-1, 7)$ and $(4, -3)$ in the ratio $2 : 3$. [2]
(A) $(1, 3)$
(B) $(2, 4)$
(C) $(3, 1)$
(D) $(-1, 2)$
- Q10.** If $\sin \theta = 1/2$, then the value of $(3 \cos \theta - 4 \cos^3 \theta)$ is: [1]
(A) 0
(B) 1
(C) $1/2$
(D) $\sqrt{3}/2$
- Q11.** If $\sin A = 1/2$, then the value of $\cot A$ is: [1]
(A) $\sqrt{3}$
(B) $1/\sqrt{3}$
(C) $1/2$
(D) 1
- Q12.** If $\sin \theta = 1/3$, then the value of $(2 \cot^2 \theta + 2)$ is: [1]
(A) 16
(B) 18
(C) 20
(D) 12
- Q13.** The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has: [1]
(A) A unique solution
(B) Exactly two solutions
(C) Infinitely many solutions
(D) No solution
- Q14.** If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then the value of k is: [1]
(A) $-5/4$
(B) $2/5$
(C) $15/4$
(D) $3/2$
- Q15.** The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ has: [1]
(A) A unique solution
(B) Exactly two solutions
(C) Infinitely many solutions
(D) No solution
- Q16.** If $x = a, y = b$ is the solution of the equations $x - y = 2$ and $x + y = 4$, then the values of a and b are respectively: [1]
(A) 3 and 5
(B) 5 and 3
(C) 3 and 1
(D) -1 and -3
- Q17.** If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is: [1]
(A) 10
(B) -10
(C) -7
(D) -2

- Q18.** A quadratic polynomial whose zeroes are -4 and 3 is: [1]
 (A) $x^2 + x - 12$
 (B) $x^2 - x + 12$
 (C) $x^2 - x - 12$
 (D) $x^2 + 7x + 12$
- Q19.** If $P(E) = 0.05$, what is the probability of 'not E'? [1]
 (A) 95
 (B) 05
 (C) 05
 (D) 50
- Q20.** Which of the following cannot be the probability of an event? [1]
 (A) $\frac{2}{3}$
 (B) -1.5
 (C) 15%
 (D) 7
- Q21.** Which of the following quadratic equations has no real roots? [1]
 (A) $2x^2 - 3x + 1 = 0$
 (B) $x^2 - 4x + 4 = 0$
 (C) $x^2 + 2x + 5 = 0$
 (D) $2x^2 - 6x + 3 = 0$
- Q22.** The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is: [1]
 (A) -8
 (B) 10
 (C) -16
 (D) 8
- Q23.** The discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ is: [1]
 (A) -8
 (B) 10
 (C) -16
 (D) 8
- Q24.** If the equation $9x^2 + 6kx + 4 = 0$ has real and equal roots, then the value of k is: [1]
 (A) 2 or -2
 (B) 4 or -4
 (C) 9 or -9
 (D) 0
- Q25.** If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x, y are prime numbers, then HCF(a, b) is: [1]
 (A) xy
 (B) xy^2
 (C) x^3y^3
 (D) x^2y^2
- Q26.** The least number that is divisible by all the numbers from 1 to 10 (both inclusive) is: [1]
 (A) 10
 (B) 100
 (C) 504
 (D) 2520
- Q27.** The exponent of 2 in the prime factorisation of 144 is: [1]
 (A) 4
 (B) 5
 (C) 6
 (D) 3

- Q28.** The decimal expansion of the rational number $14587 / (1250)$ will terminate after how many decimal places? [1]
- (A) 1
(B) 2
(C) 3
(D) 4
- Q29.** The angle of elevation of the top of a tower from a point on the ground, 30 m away from the foot of the tower, is 30 deg. [1]
The height of the tower is:
- (A) $10\sqrt{3}$ m
(B) $30\sqrt{3}$ m
(C) $20\sqrt{3}$ m
(D) 10 m
- Q30.** The angle of elevation of the top of a 30 m high tower from a point on the ground is 30 deg. The distance of the point [1]
from the base of the tower is:
- (A) 30 m
(B) $30\sqrt{3}$ m
(C) $10\sqrt{3}$ m
(D) 60 m
- Q31.** If the shadow of a tower is $\sqrt{3}$ times its height, the sun's angle of elevation is: [2]
- (A) 30 deg
(B) 45 deg
(C) 60 deg
(D) 90 deg
- Q32.** The empirical relationship between the three measures of central tendency is: [1]
- (A) $3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$
(B) $\text{Mode} = 3 \text{ Mean} - 2 \text{ Median}$
(C) $2 \text{ Median} = 3 \text{ Mode} - \text{Mean}$
(D) $\text{Mean} = 3 \text{ Median} - 2 \text{ Mode}$
- Q33.** The empirical relationship between the three measures of central tendency is: [1]
- (A) $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$
(B) $\text{Mode} = 2 \text{ Median} - 3 \text{ Mean}$
(C) $\text{Median} = 3 \text{ Mode} - 2 \text{ Mean}$
(D) $\text{Mean} = 3 \text{ Median} - 2 \text{ Mode}$
- Q34.** For a given distribution, Mode = 24 and Mean = 30. The value of Median using the empirical relationship is: [1]
- (A) 26
(B) 28
(C) 27
(D) 25
- Q35.** The modal class for the following frequency distribution: Classes: 0-10 (f=5), 10-20 (f=8), 20-30 (f=15), 30-40 (f=10), [1]
40-50 (f=6) is:
- (A) 20
(B) 30
(C) 40
(D) 50
- Q36.** If the radius of the base of a cone is halved, keeping the height same, what is the ratio of the volume of the reduced cone [1]
to that of the original cone?
- (A) 1 : 4
(B) 1 : 2
(C) 1 : 8
(D) 1 : 16

SECTION B - SHORT ANSWER QUESTIONS

(13 Questions * 40 Marks)

- Q37.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes. (use $\pi = 22/7$) [3]
- Q38.** If the sum of the first 14 terms of an AP is 1050 and its first term is 10, find the 20th term. [3]
- Q39.** If $\sec \theta + \tan \theta = p$, prove that $\sin \theta = (p^2 - 1)/(p^2 + 1)$. [4]
- Q40.** Solve the following pair of linear equations by elimination method: $3x + 4y = 10$ and $2x - 2y = 2$. [3]
- Q41.** Find the zeroes of the quadratic polynomial $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients. [3]
- Q42.** If the squared difference of the zeroes of the quadratic polynomial $f(x) = x^2 + px + 45$ is equal to 144, find the value of p . [3]
- Q43.** Solve the quadratic equation for x : $4x^2 - 4ax + (a^2 - b^2) = 0$ using the quadratic formula. [3]
- Q44.** Find the HCF and LCM of 404 and 96 by prime factorisation method and verify that $\text{HCF} \times \text{LCM} = \text{Product of the two given numbers}$. [3]
- Q45.** From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30 deg and 45 deg respectively. If the bridge is at a height of 3 m from the banks, find the width of the river. [3]
- Q46.** The mean of the following frequency distribution is 18. Find the missing frequency f :
Class: 11-13, 13-15, 15-17, 17-19, 19-21, 21-23, 23-25
Frequency: 7, 6, 9, 13, f , 5, 4 [3]
- Q47.** A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel. (Take $\pi = 22/7$) [3]
- Q48.** A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m, find the area of the canvas used for making the tent. (Take $\pi = 22/7$) [3]
- Q49.** In ΔABC , D is a point on side BC such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \times CD$. [3]

SECTION C - LONG ANSWER QUESTIONS

(4 Questions * 20 Marks)

- Q50.** Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre. [5]
- Q51.** If $A(-5, 7)$, $B(-4, -5)$, $C(-1, -6)$ and $D(4, 5)$ are the vertices of a quadrilateral, find the area of the quadrilateral $ABCD$. [5]
- Q52.** A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $2/3$. Find the number of blue marbles in the jar. If 6 more blue marbles are added to the jar, find the new probability of drawing a blue marble. [5]
- Q53.** BL and CM are medians of a triangle ABC right-angled at A . Prove that $4(BL^2 + CM^2) = 5 BC^2$. [5]

*** END OF QUESTION PAPER ***